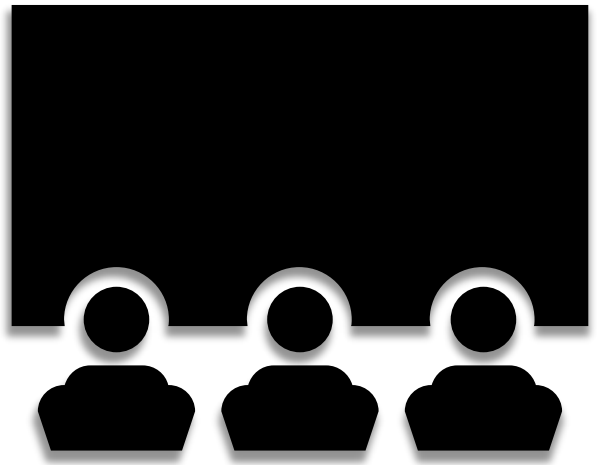
Machine Learning Individual Project

SARAH WOODCOCK OCTOBER 2026

Comparative Intelligence and Ethics in Machine Learning Systems



Dataset and Track Selection

- Project Track:
  - I chose Track B: Recommender Systems (RS)

- Dataset:
  - I chose MovieLens 1M
  - https://grouplens.org/datasets/movielens/1m/

There were Pros and Cons to this dataset choice

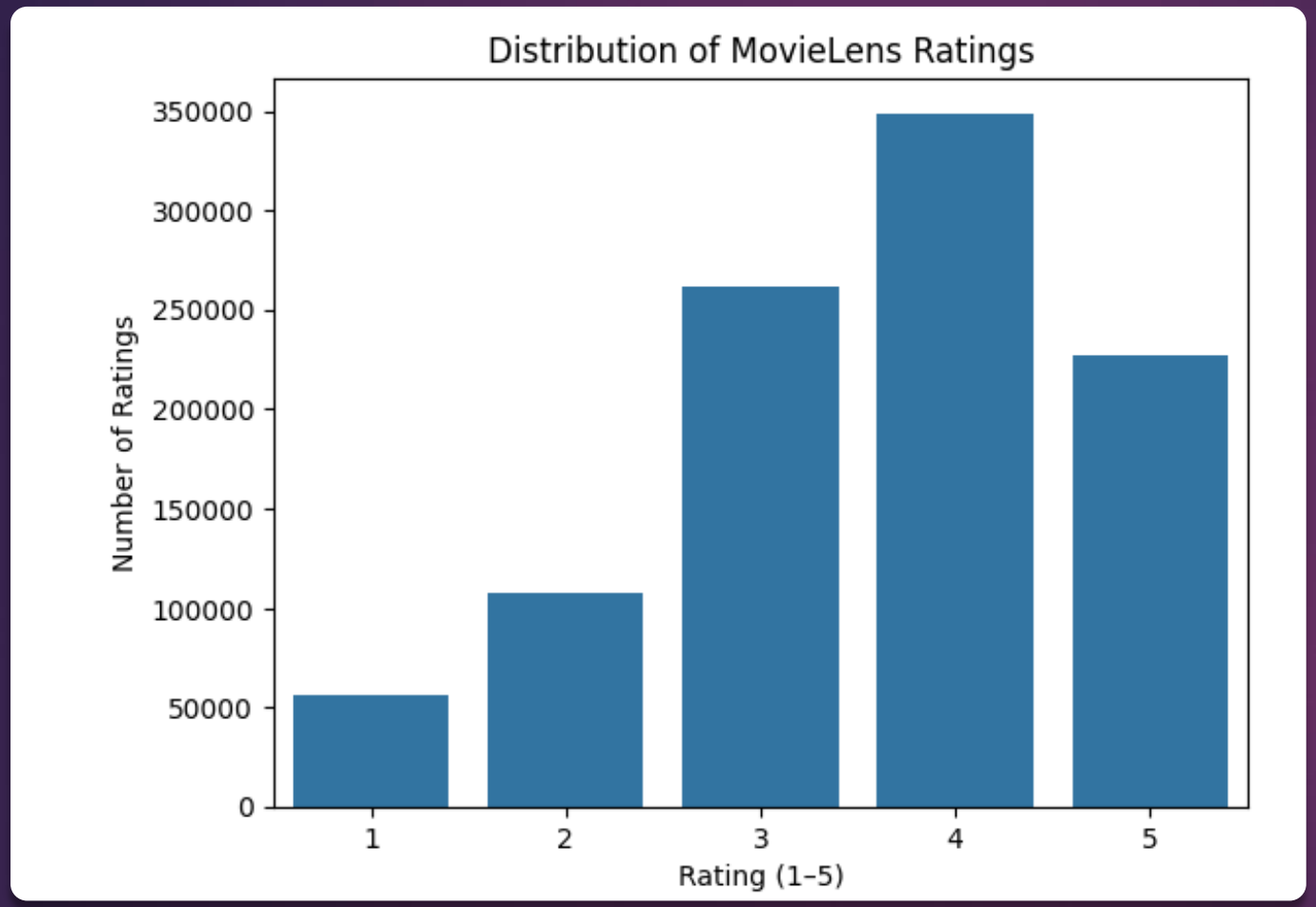
Pros

- A very large amount of data with many facets – information about the films and the users who rated them was well-rounded
- The dataset from MovieLens 100k was used in our Unit08 example (University of Essex Online 2026). I already had some information I could compare my results with to check I was on the right track.
- In comparison with other datasets I considered – Book Crossing and some theatrical review datasets - MovieLens 1M held much more data per item and per user, giving more facets to explore

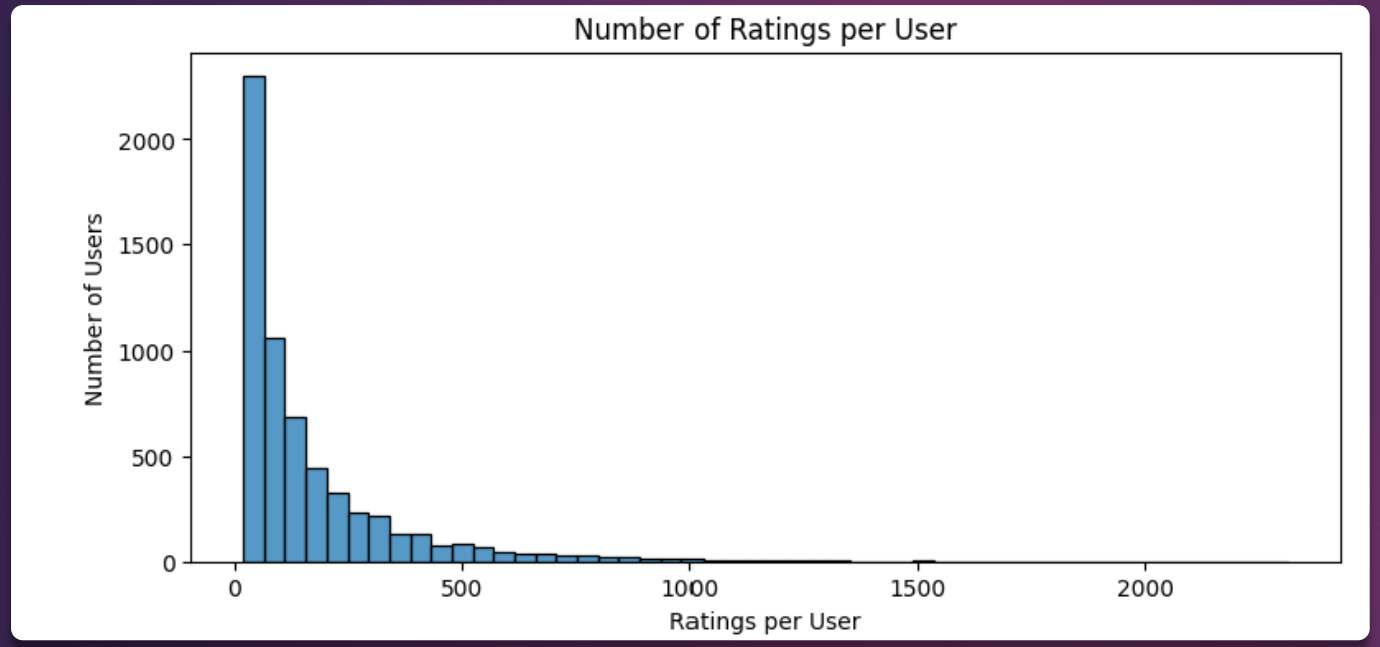
Cons

- The dataset from MovieLens 100k was used in our Unit08 example. It would be difficult to write code that hadn't already been used.
- The dataset is well known and widely used so it would be difficult to draw new conclusions.

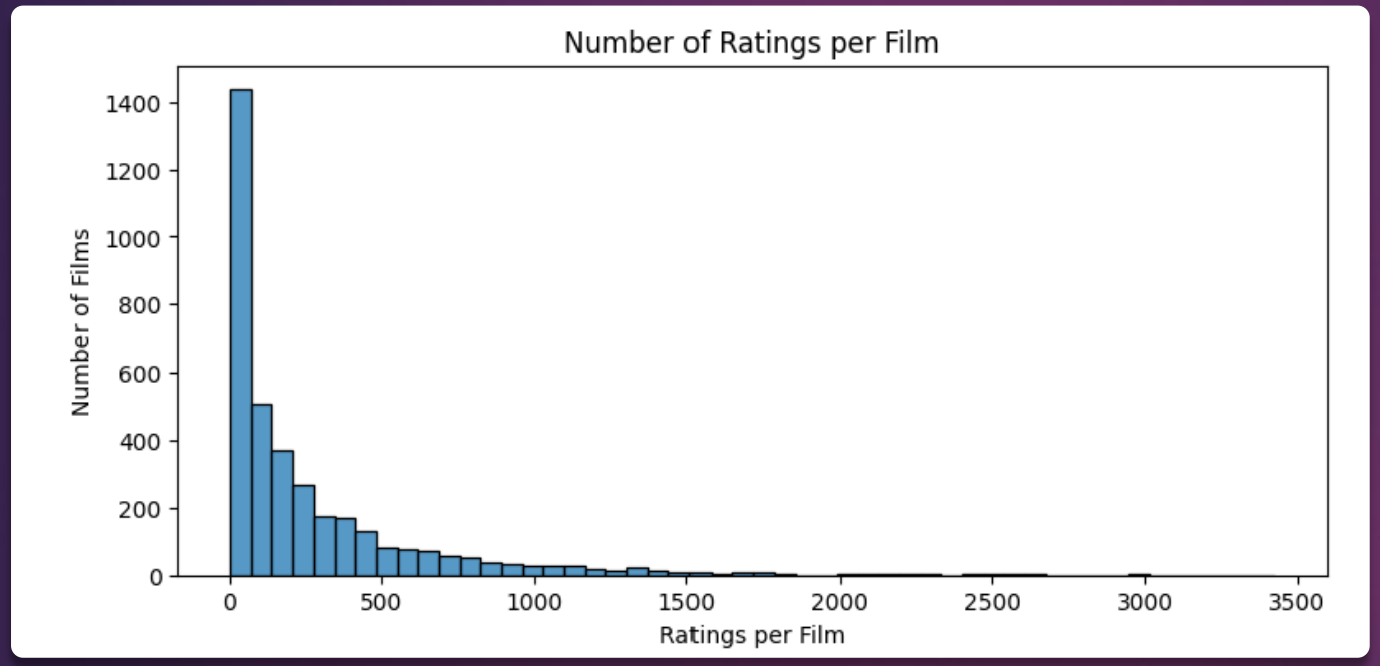
Exploratory Data Analysis – Ratings Distribution



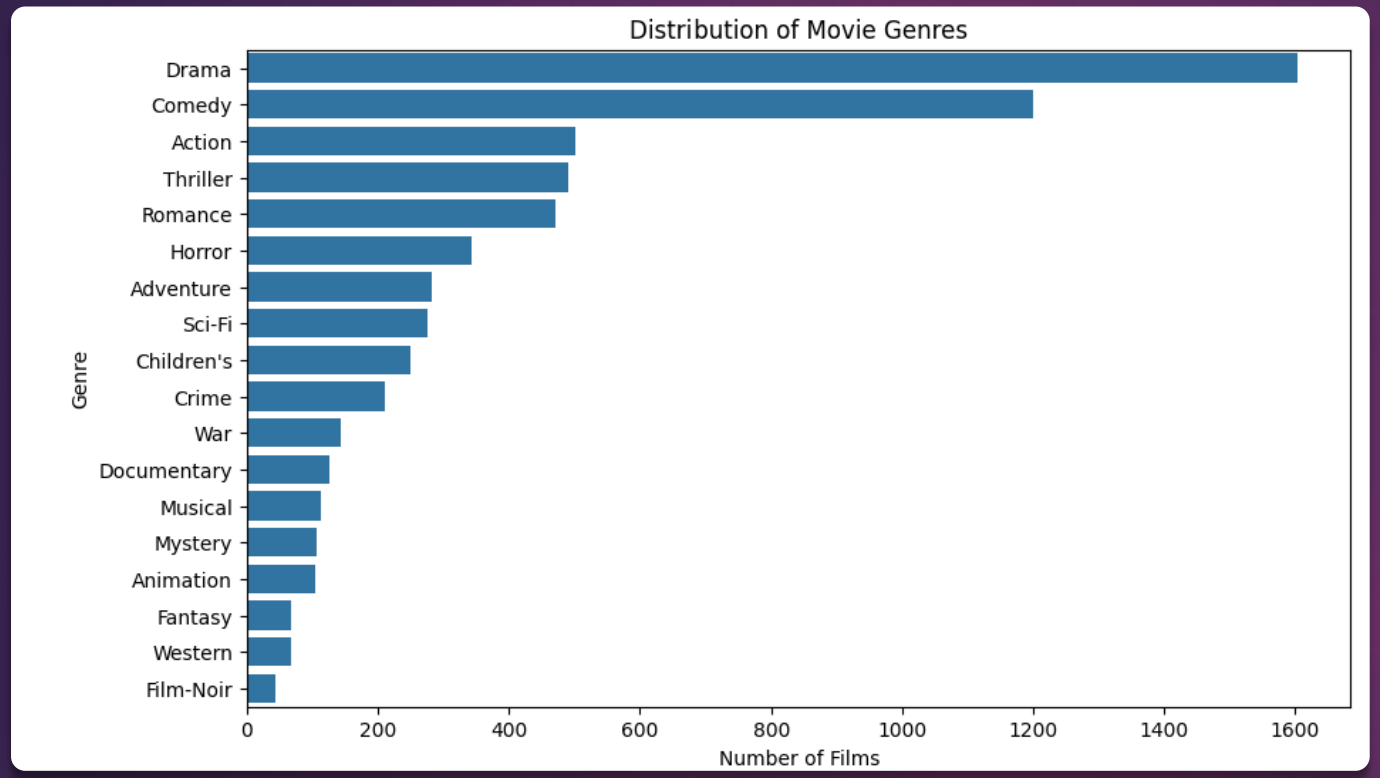
Exploratory Data Analysis - Ratings per user

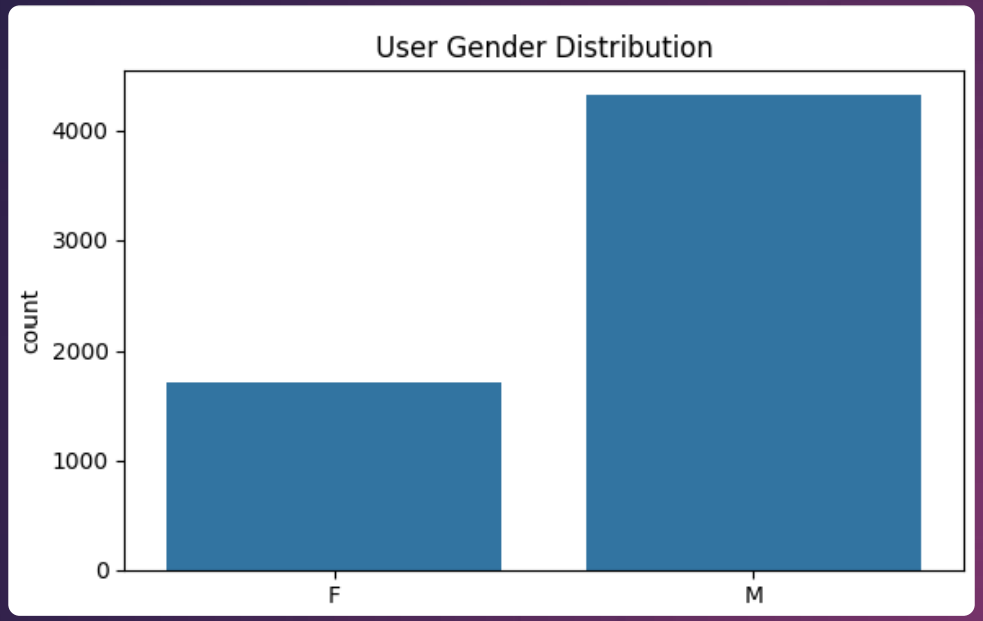


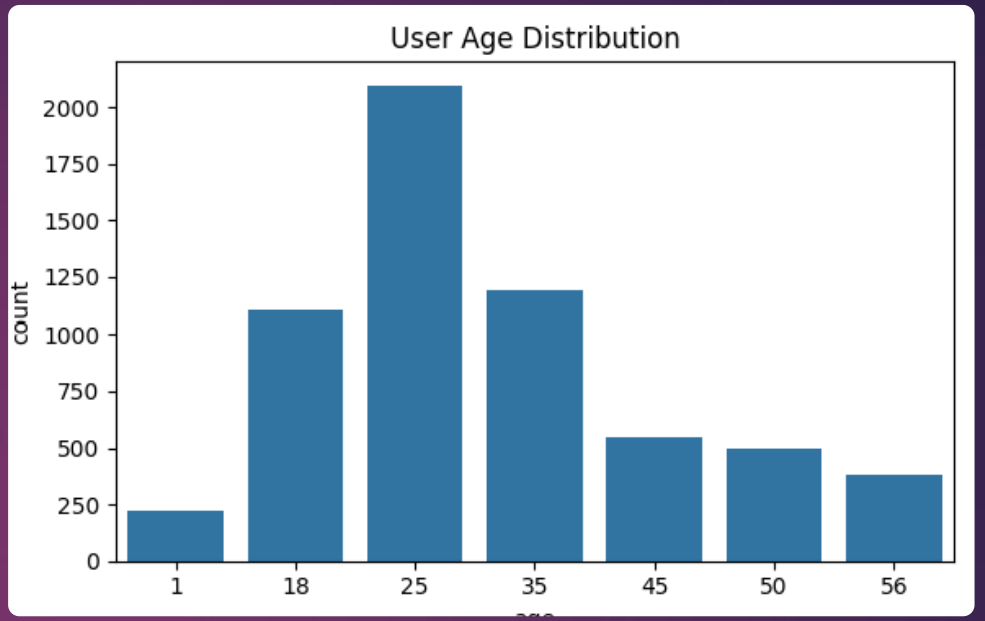
Exploratory Data Analysis – Ratings per film



Exploratory Data Analysis – Film Genres





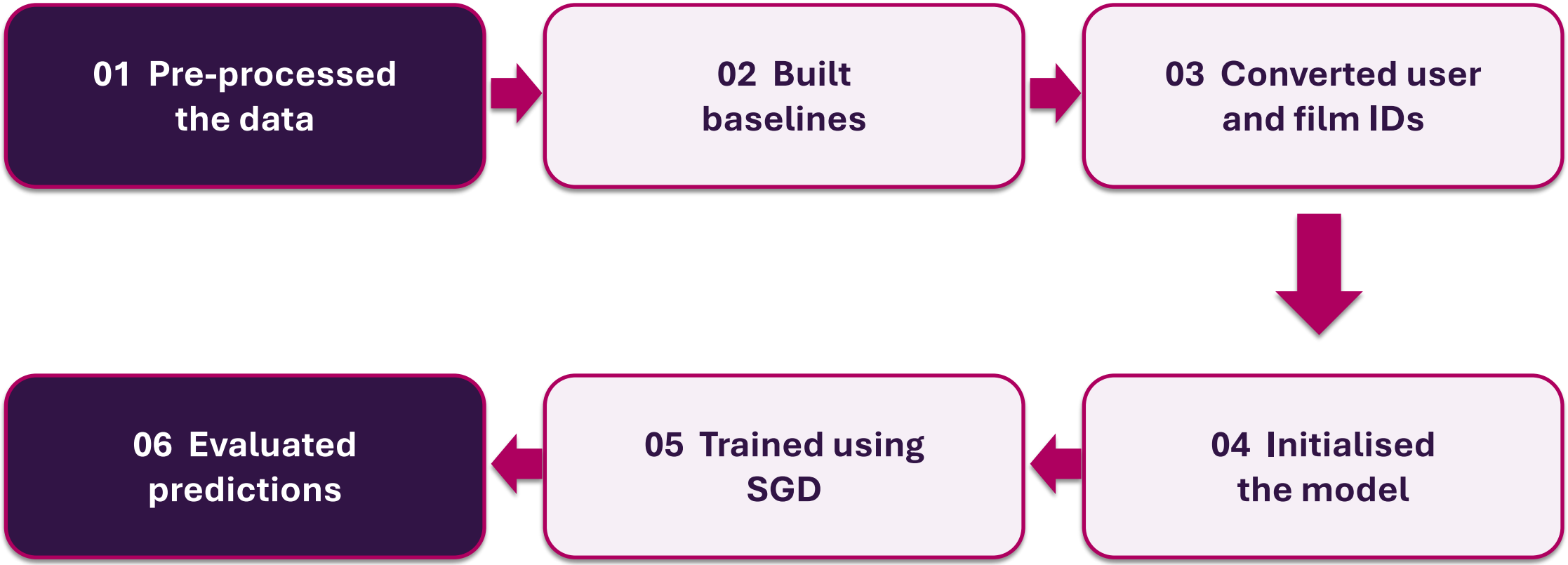


Exploratory Data Analysis – User Demographics

Traditional Machine Learning Model

MODEL-BASED COLLABORATIVE FILTERING USING MATRIX FACTORISATION, TRAINED WITH STOCHASTIC GRADIENT DESCENT (SGD) – BASED ON UNIT08 EXERCISE

Steps used to build the traditional model



Data Preprocessing

- I imported the data
- I validated the data – e.g. checking all ratings were between 1 and 5
- I carried out Exploratory Data Analysis
- I checked the numbers of ratings, users and films and calculated sparsity (95.5%) 3,706 films. 1,000,209 ratings. 6,040 users
- I sorted the ratings by timestamp and split the data 80/10/10 (Training/Validation/Test)

Baselines

Model	Dataset	RMSE	MAE
Global mean	Validation	1.1191	0.9366
Global mean	Test	1.0893	0.9025
Bias baseline	Validation	0.9726	0.7759
Bias baseline	Test	0.9451	0.7411

- Global Mean – predicts the same (mean average) rating for every user and film. This is used as a benchmark
- Bias Baseline – accounts for users who always give high/low ratings. Accounts for films that usually receive high/low ratings. Results are better than global mean overall – as expected.
- Global mean and Bias Baseline show slightly better results against test data than validation data. I would consider checking for overfitting by further exploring the RMSE variations between the training and validation datasets. The results could also be explained by the different time periods covered by the data.

Matrix Factorisation using Stochastic Gradient Descent (SGD)

I used matrix factorisation trained using SGD. This gave a lower RMSE of 0.9187.

This shows that allowing the model to discover hidden features of films and users can further improve prediction accuracy.

The NDCG@10 of this model was 0.8785 (976 users evaluated)

UI Demonstration

I built a simple UI using Gradio to show the model in action. Here are some screenshots to show its performance

MovieLens Recommendation System

Classical Machine Learning: Matrix Factorisation (SGD)

Personalised Recommendations | Cold-Start Demonstration | Popularity Bias

Select User ID

647

Recommend Films

Top 5 Recommended Films

Movie	Predicted Rating	Number of Ratings
Boys Don't Cry (1999)	5.01	687
Fargo (1996)	4.92	2144
Monty Python and the Holy Grail (1974)	4.9	1263
Blade Runner (1982)	4.89	1520
400 Blows, The (Les Quatre cents coups) (1959)	4.83	150

Runs · Use via API · Built with Gradio · Settings

MovieLens Recommendation System

Classical Machine Learning: Matrix Factorisation (SGD)

Personalised Recommendations | Cold-Start Demonstration | Popularity Bias

A new user has no rating history. The system uses a non-personalised baseline.

Recommend for New User

Cold-Start Recommendations

Film	Baseline Score	Number of Ratings
Seven Samurai (The Magnificent Seven) (Shichinin no samurai) (1954)	4.56	517
Shawshank Redemption, The (1994)	4.55	1845
Sunset Blvd. (a.k.a. Sunset Boulevard) (1950)	4.51	389
Usual Suspects, The (1995)	4.51	1466
Godfather, The (1972)	4.51	1874

Runs · Use via API · Built with Gradio · Settings

MovieLens Recommendation System

Classical Machine Learning: Matrix Factorisation (SGD)

Personalised Recommendations | Cold-Start Demonstration | Popularity Bias

Select User ID

647

Compare Recommendations

Personalised Recommendations

Film	Predicted Rating	Popularity
Boys Don't Cry (1999)	5.01	687
Fargo (1996)	4.92	2144
Monty Python and the Holy Grail (1974)	4.9	1263
Blade Runner (1982)	4.89	1520
400 Blows, The (Les Quatre cents coups) (1959)	4.83	150

Most Popular Unseen Films

Movie	Popularity
Star Wars: Episode V - The Empire Strikes Back (1980)	2516
Star Wars: Episode IV - A New Hope (1977)	2516
Star Wars: Episode VI - Return of the Jedi (1983)	2457
Saving Private Ryan (1998)	2245
Silence of the Lambs, The (1991)	2151

Popularity Comparison

0 of the 5 personalised recommendations also appear in the 5 most popular unseen movies. Average popularity of personalised recommendations: 1152.8 training ratings.

Runs · Use via API · Built with Gradio · Settings

Deep learning Model

NEURAL COLLABORATIVE FILTERING ARCHITECTURE (NCF)

Building and Training the NCF model

I reused the MovieLens 1M dataset, chronological 80/10/10 split and numerical user/film IDs.

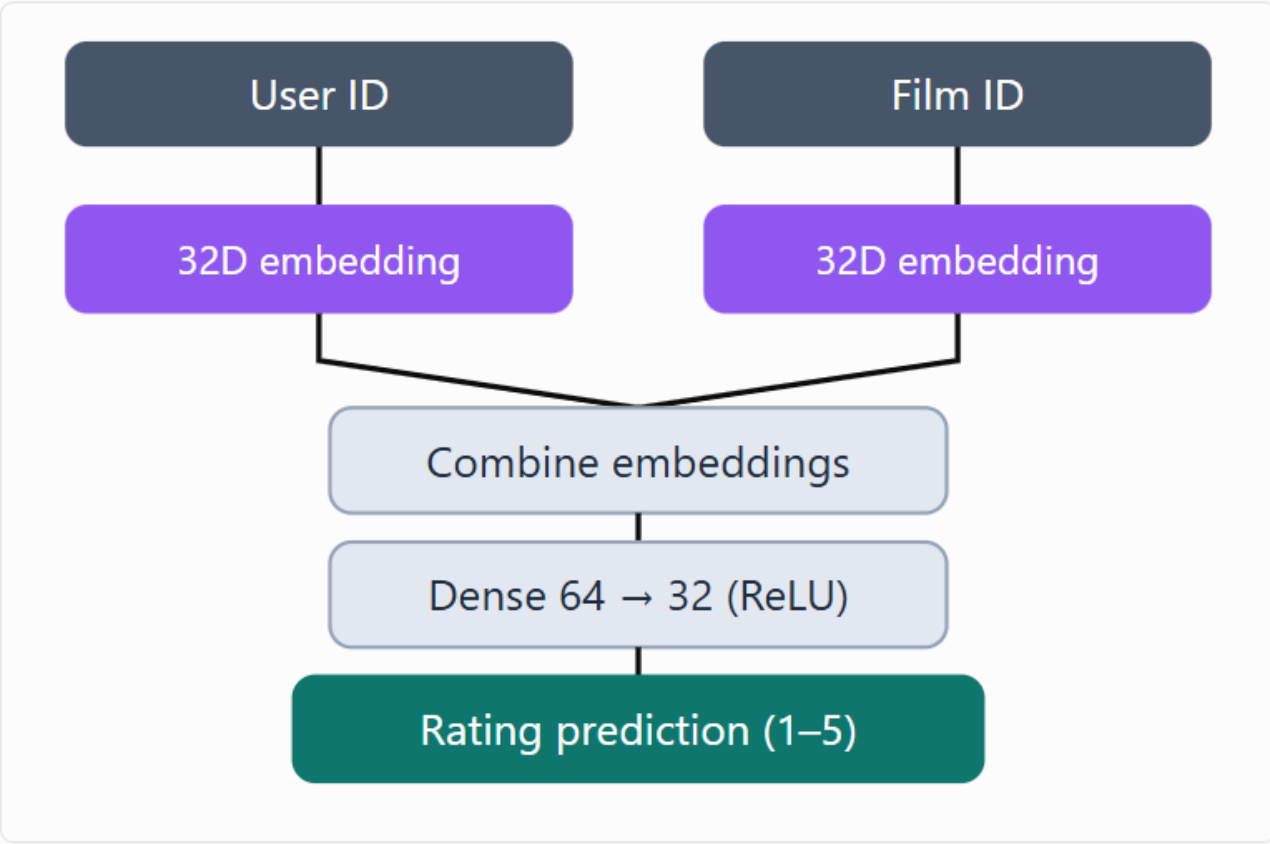
I created embedding layers with 32 dimensions for each user and film.

I combined the embeddings and passed them through two dense layers containing 64 and 32 neurons, using ReLU activation.

I used a sigmoid output, rescaled to predict ratings between 1 and 5.

I compiled the model using the Adam optimiser (learning rate 0.001) and Mean Squared Error loss.

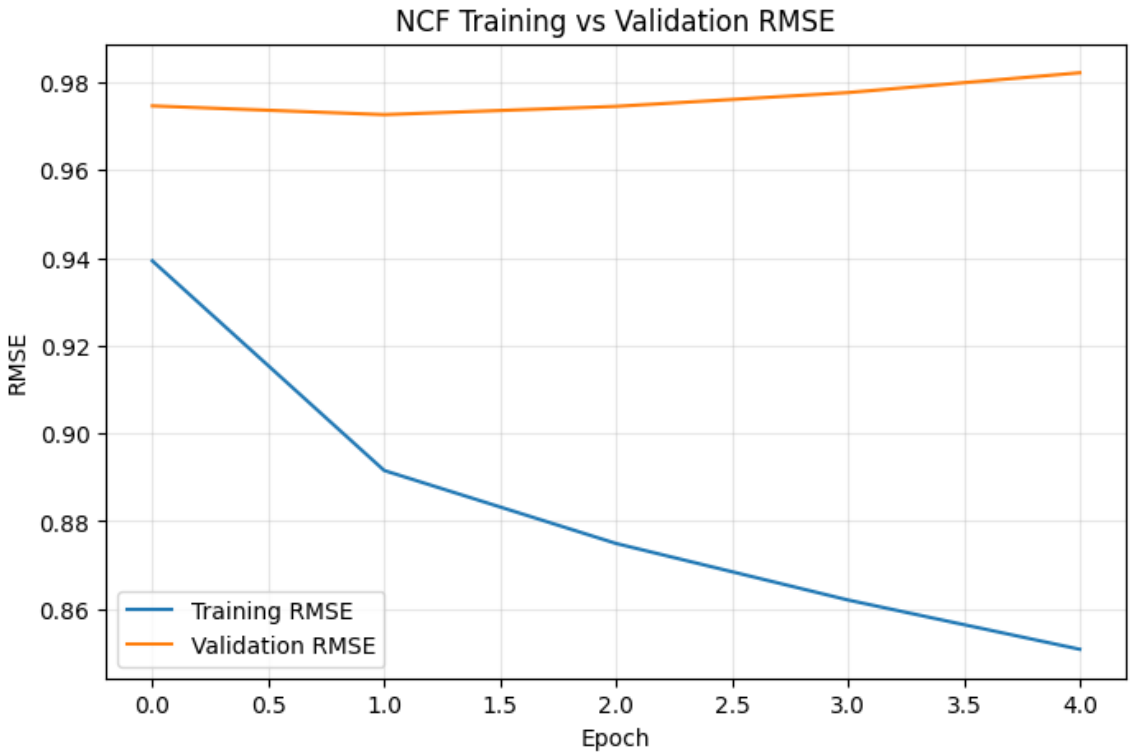
I trained the model with a maximum of 20 epochs, using early stopping to reduce the risk of overfitting.

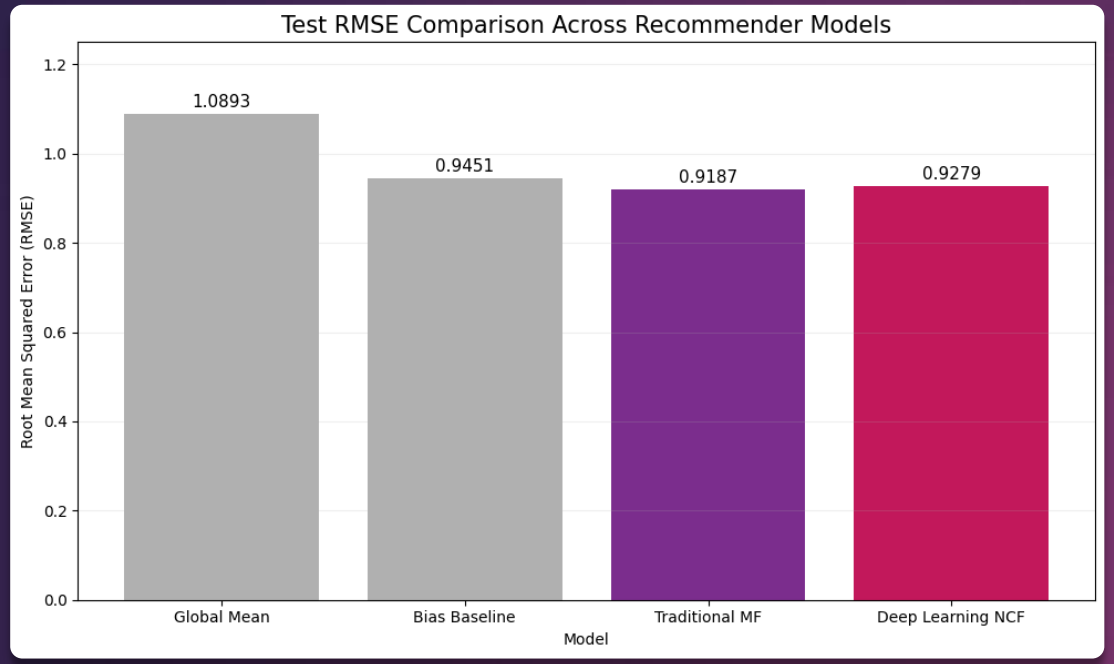


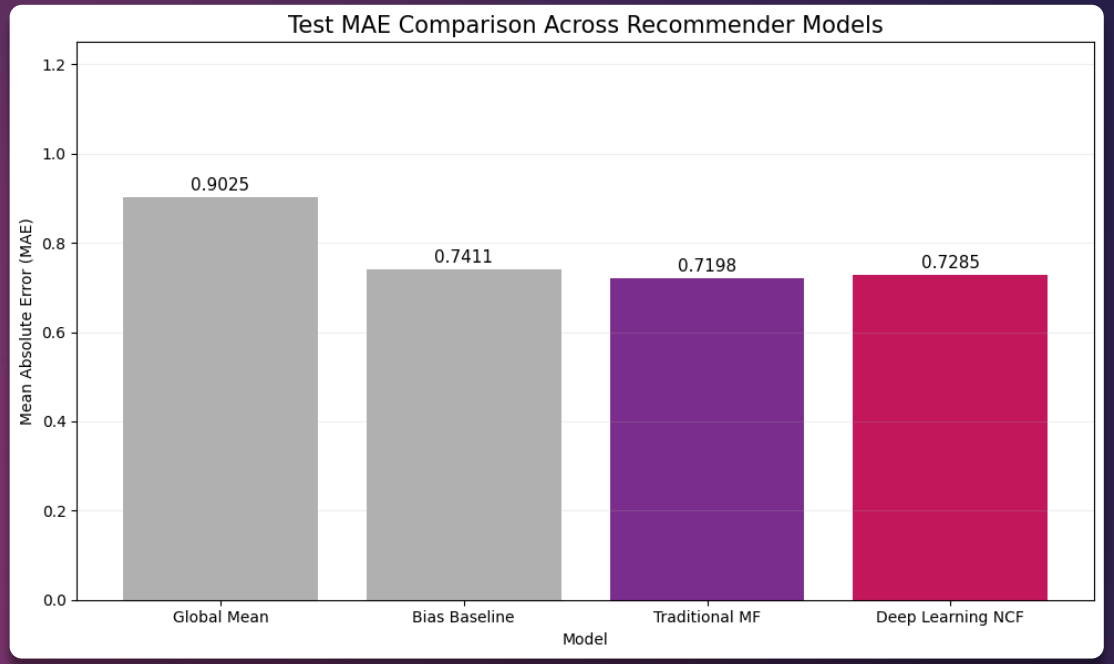
NCF Architecture

Model Evaluation

Model	Dataset	RMSE	MAE
NCF	Test	0.9279	0.7285

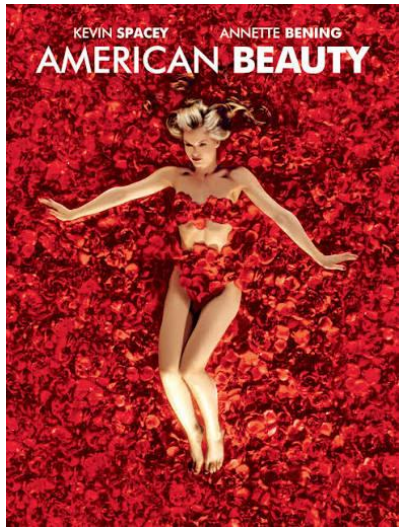






Test RMSE and MAE by model

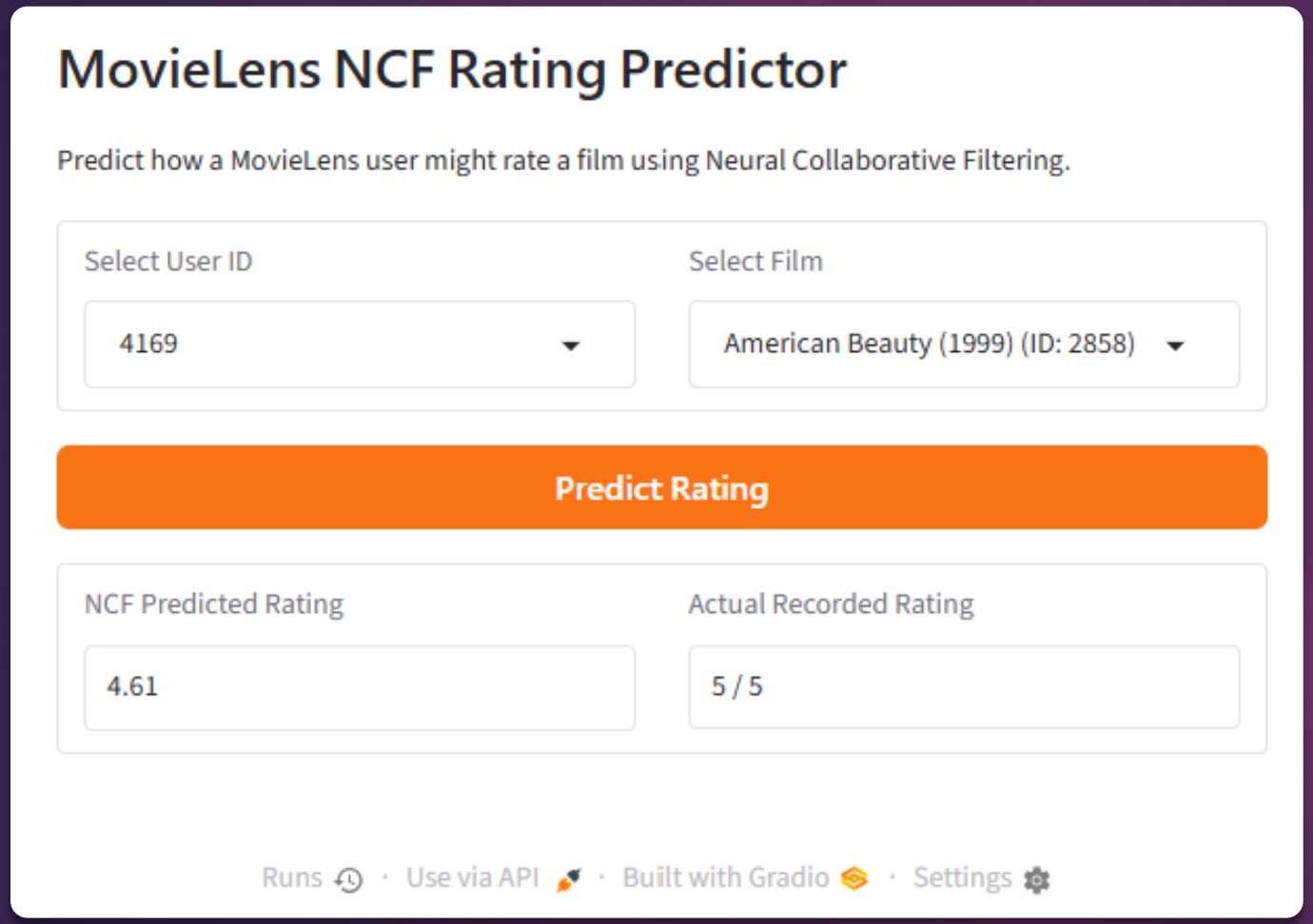
NDCG@10 – MF: 0.8785. NCF: 0.8769 across the same 976 users



- Most rated film: American Beauty (1999) had 3,428 ratings
- User with most ratings – User 4169 with 2,314 (Male, age 50)
- Did he watch American Beauty? Yes – and he rated it 5/5.



- Least rated film: Juno and Paycock (1930) had 1 rating
- User with least ratings – User 5725 with 20 ratings (Male, age 18)
- Did he watch Juno and the Paycock? No.
- The user who rated Juno and the Paycock was user 4802 (Male, age 56)



Gradio UI showing a prediction for a specific user

Critical evaluation: MF vs NCF

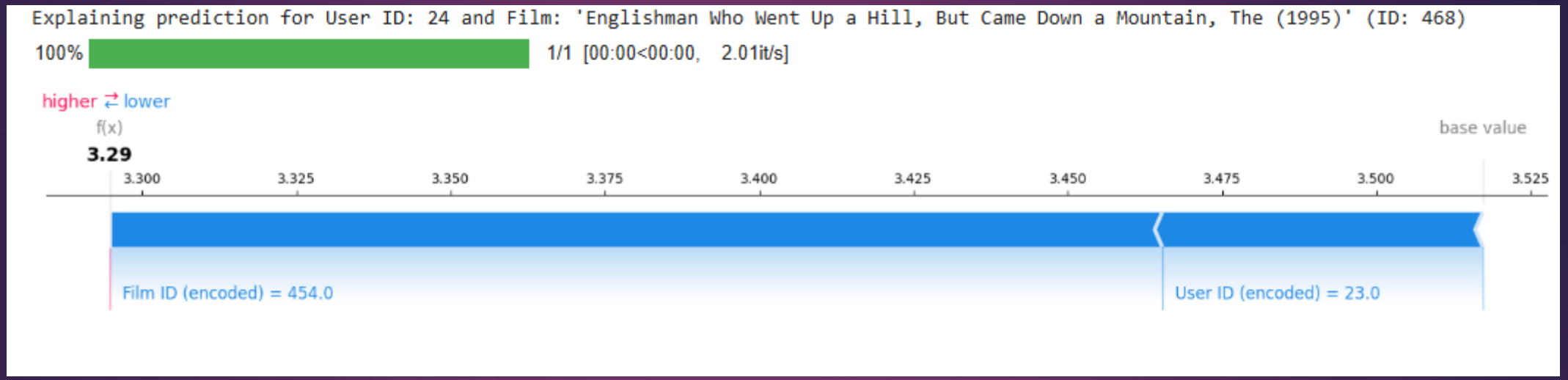
Comparison	Matrix Factorisation (MF)	Neural Collaborative Filtering (NCF)
Architecture	Relatively simple	More complex
Prediction accuracy	Lowest RMSE and MAE in this experiment	Did not outperform MF
Learning capability	Limited nonlinear modelling	Learns nonlinear relationships
Explainability	User/film biases partly interpretable; latent factors less so	Embeddings and hidden layers difficult to interpret
Training	Fewer architectural decisions	More hyperparameter tuning complexity; flexible training
Deployment	Easier to maintain and deploy	Greater complexity
Cold-start and sparsity	Requires historical ratings; popularity bias risk	Also affected by sparse ratings and cold-start

Explainability

```
Surrogate fidelity RMSE (surrogate vs MF): 0.005968734978210198

Explanation for: User 24 -> Movie 468 (Englishman Who Went Up a Hill, But Came Down a Mountain, The (1995))
MF prediction:        3.214
Surrogate prediction: 3.217
SHAP base value:      3.635
Top SHAP contributions:
          item_bias: -0.3065
          user_bias: +0.0917
   latent_inter_29: -0.0439
    latent_inter_3: -0.0388
   latent_inter_10: -0.0323
   latent_inter_18: +0.0323
   latent_inter_19: +0.0251
    latent_inter_2: -0.0247
    latent_inter_0: -0.0211
   latent_inter_21: +0.0189
```

Explainability Findings – SHAP - MF



Explainability Findings – SHAP - NCF

Ethical and Bias Considerations

Ethical Considerations



Exploratory data analysis showed uneven distributions of ratings across users and films.

There is a risk that models will learn the most from frequently rated films and highly active users.

This could put less popular films at a disadvantage and provide less helpful recommendations to users with limited rating histories.

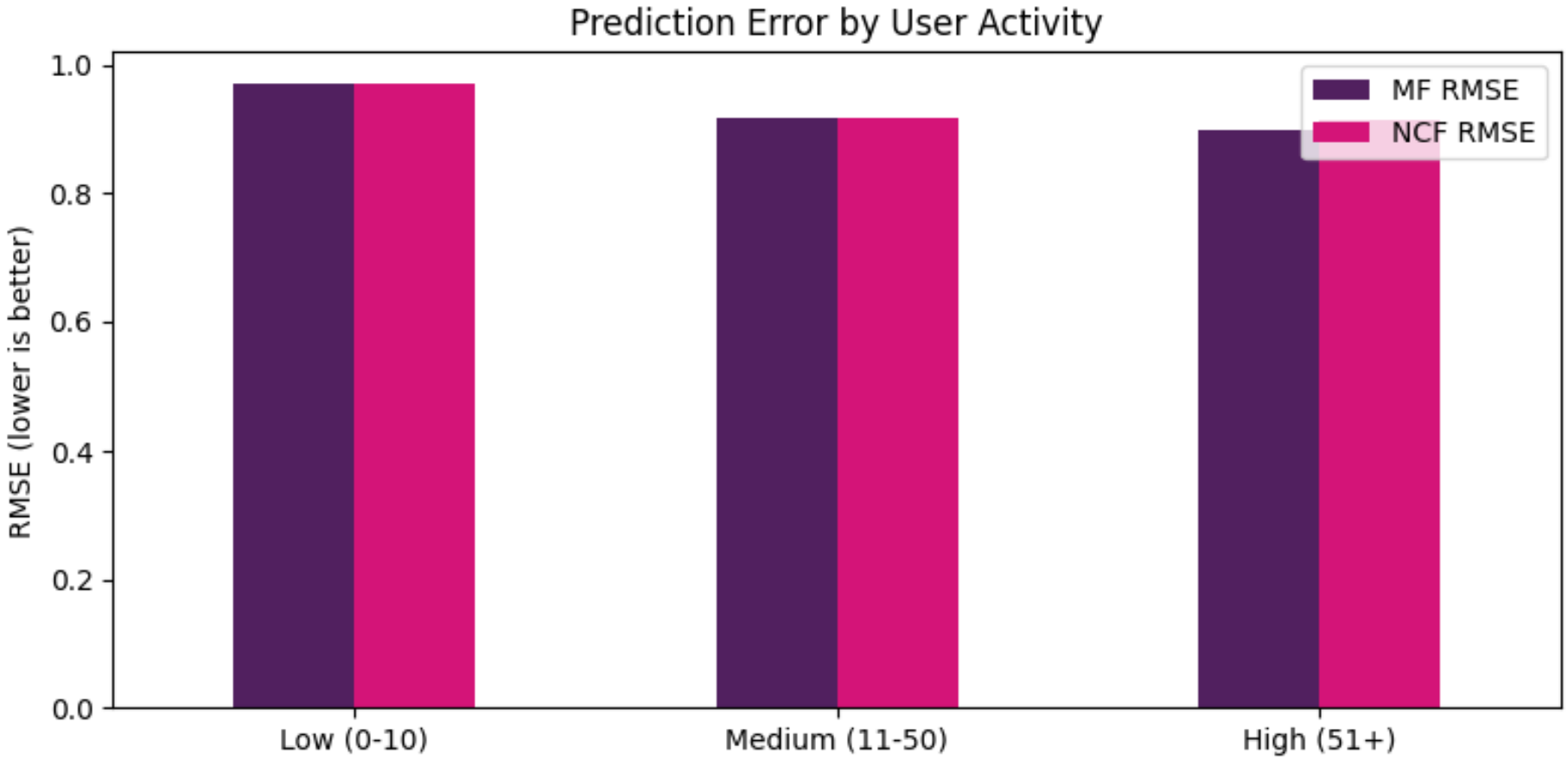
MovieLens 1M contains demographic information, allowing the representation of different user groups to be examined and considered within our recommender system.

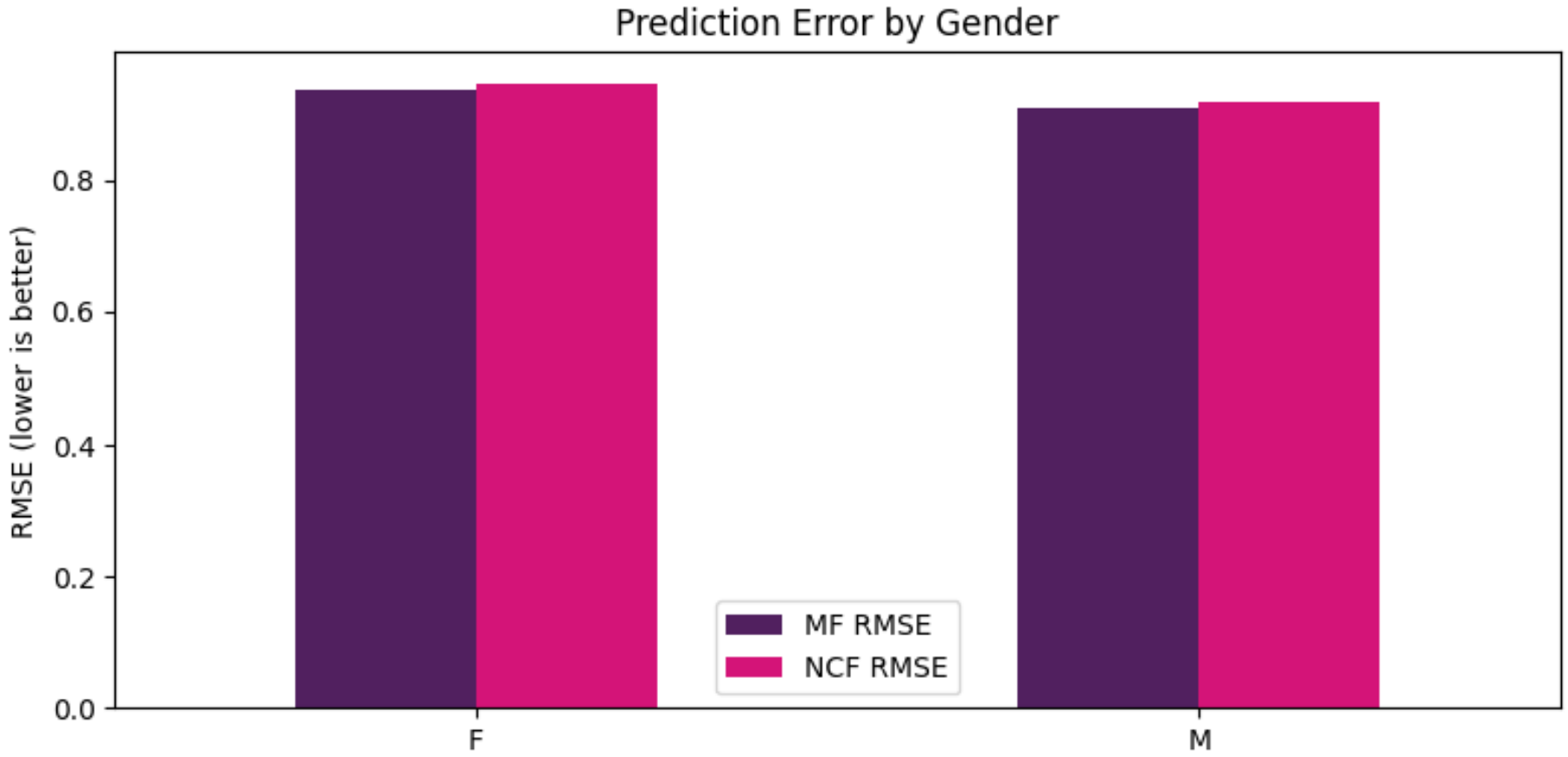
However, the above points raise questions about fairness, representation and whether the dataset reflects the wider population of potential users.

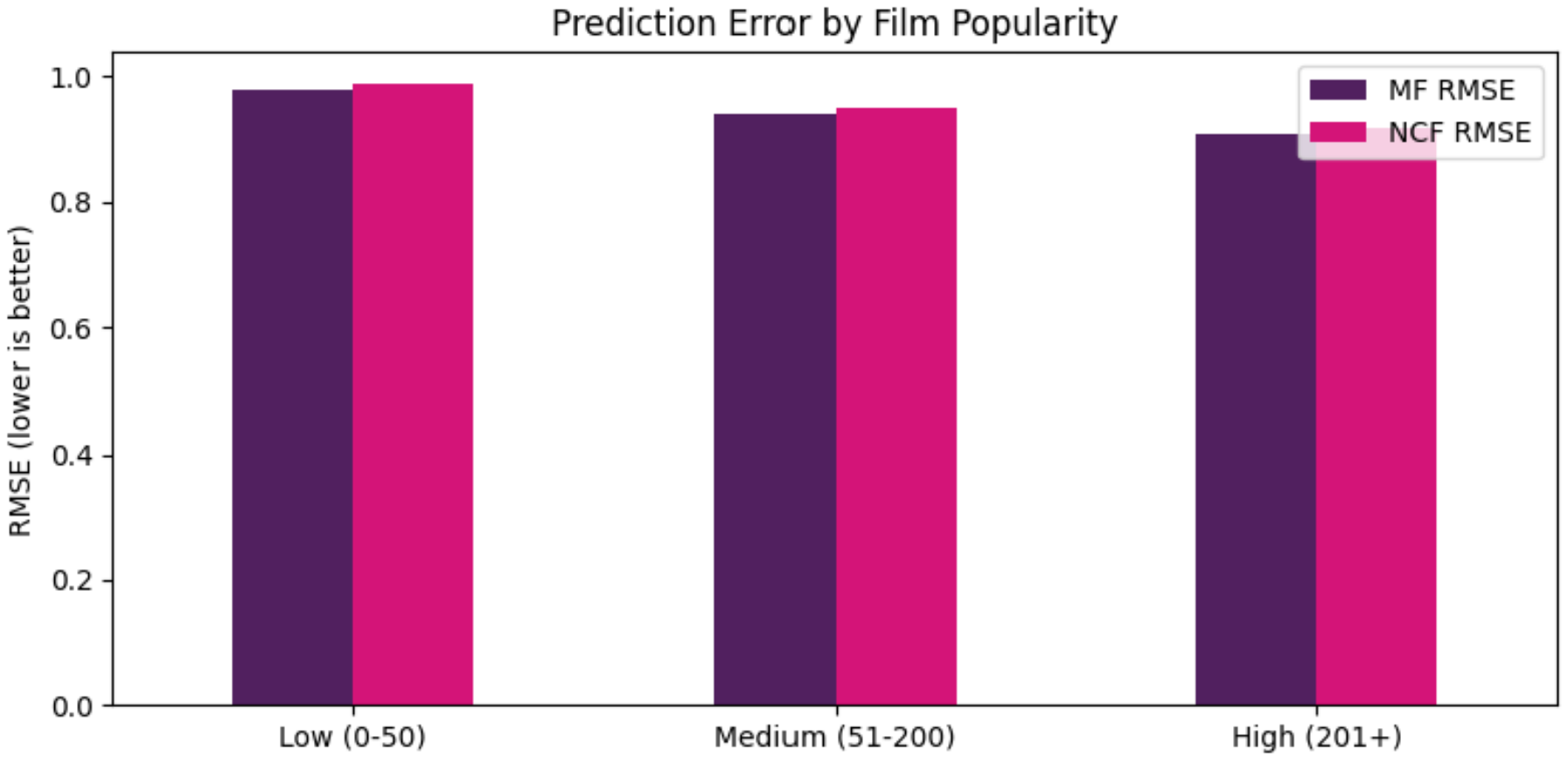
Key biases in the MovieLens data



- Demographic – different ages and genders were differently represented.
- Popularity – some films were rated far more times than others
- Activity – some users were extremely active and so contributed more data


Prediction Error by User Activity
RMSE (lower is better)
1.0
0.8
0.6
0.4
0.2
0.0
MF RMSE
NCF RMSE
Low (0-10)
Medium (11-50)
High (51+)


Prediction Error by Gender
RMSE (lower is better)
0.0
0.2
0.4
0.6
0.8
F
M
MF RMSE
NCF RMSE


Prediction Error by Film Popularity
RMSE (lower is better)
1.0
0.8
0.6
0.4
0.2
0.0
MF RMSE
NCF RMSE
Low (0-50)
Medium (51-200)
High (201+)

Deployment Considerations

- Machine Learning uses a lot of resources to run. Implementing the simpler model will be less of a drain on these resources. As NCF did not show any real improvement, I would choose MF based on the current findings.
- Cold-start users and films (those about which we hold no ratings) will always be a problem as we don't know what to base our recommendations on. A real-world implementation would need a measure to address this, such as a Top 5 films question for a new user.
- Either system would require ongoing monitoring to ensure reasonable predictions were being made. Low activity users, female users and films with lower ratings frequency had the highest error rates. We'd want to improve this over time.
- Although it's anonymised here, personal data is held regarding the users and this would need to be carefully and sensitively stored, subject to UK GDPR and other relevant rules across other countries.
- New films are released all the time, so the database would need to be regularly updated and the model retrained with the new data. This maintenance would be easier on the traditional model.



Conclusion

In the case of MovieLens 1M, I'd be inclined to use a traditional matrix factorisation model for my recommender system. This is for the following reasons:

- It's easier to build and maintain than an NCF model
- It achieved the lowest test RMSE (0.9187) in my results
- It achieved the lowest test MAE (0.7198) in my results
- It achieved the highest NDCG@10 (0.8785) in my results
- User and film biases are accounted for, making the recommendations more understandable.
- NCF did not give improved results, despite its additional complexity, although the results were extremely close.

References

- Chat GPT – feedback on how my presentation fitted the submission brief. https://chatgpt.com/share/6acbb412-ea84-83ed-9a67-20ad1bb1185e
- Gemini – in Google Colab
- Géron, A. (2025) Hands-on machine learning with Scikit-Learn and PyTorch: concepts, tools, and techniques to build intelligent systems. Sebastopol, CA: O'Reilly Media.
- He, X., Liao, L., Zhang, H., Nie, L., Hu, X. and Chua, T.-S. (2017) 'Neural collaborative filtering', Proceedings of the 26th International Conference on World Wide Web, pp. 173–182. Available at: https://doi.org/10.1145/3038912.3052569 (Accessed: 11 October 2026).
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